

Evaluate the following integrals.

$$1). \int \sin^3 x \cos^2 x \, dx \quad u = \cos x \\ du = -\sin x \, dx$$

$$= \int \sin^2 x \cos^2 x \sin x \, dx$$

$$= \int (1 - \cos^2 x) \cos^2 x \sin x \, dx$$

$$= \int (1 - u^2) u^2 \cdot (-du)$$

$$= -\int (u^2 - u^4) \, du$$

$$= -\int (u^2 - u^4) \, du$$

$$= -\frac{1}{3} u^3 + \frac{1}{5} u^5 + C$$

$$= -\frac{1}{3} \cos^3 x + \frac{1}{5} \cos^5 x + C$$

$$2). \int_0^{\pi/6} \cos^5 x \, dx$$

$$u = \sin x \quad du = \cos x \, dx$$

$$\text{When } x=0, \quad u=0$$

$$\text{When } x=\frac{\pi}{6}, \quad u=\frac{1}{2}$$

$$= \int_0^{\pi/6} \cos^4 x \cos x \, dx$$

$$= \int_0^{\pi/6} (1 - 2\sin^2 x + \sin^4 x) \cos x \, dx \quad \cos^4 x = (\cos^2 x)^2 \\ = (1 - \sin^2 x)^2$$

$$= \int_0^{1/2} (1 - 2u^2 + u^4) \, du \quad = 1 - 2\sin^2 x + \sin^4 x$$

$$= \left(u - \frac{2}{3} u^3 + \frac{1}{5} u^5 \right) \Big|_0^{1/2} \quad = 1 - 2u^2 + u^4$$

$$= \frac{1}{2} - \frac{2}{3} \cdot \left(\frac{1}{2} \right)^3 + \frac{1}{5} \left(\frac{1}{2} \right)^5$$

$$= \frac{1}{2} - \frac{1}{12} + \frac{1}{160} = \frac{203}{480}$$

$$3). \int_0^{\pi/2} \cos^2 \theta \, d\theta$$

$$= \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} \, d\theta$$

$$= \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2} \cos 2\theta \right) d\theta$$

$$= \left(\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{\pi/2}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) + \frac{1}{4} (\sin \pi - \sin 0)$$

$$= \frac{\pi}{4}$$

$$4). \int \sec^2 x \tan x \, dx$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$= \int u \, du$$

$$= \frac{1}{2} u^2 + C$$

$$= \frac{1}{2} \tan^2 x + C$$

$$5). \int \tan^3 x \sec x \, dx$$

$$= \int \tan^2 x \cdot \tan x \sec x \, dx$$

$$= \int (\sec^2 x - 1) \tan x \sec x \, dx$$

$$= \int (u^2 - 1) \, du \quad \begin{array}{l} u = \sec x \\ du = \sec x \tan x \, dx \end{array}$$

$$= \frac{1}{3} u^3 - u + C$$

$$= \frac{1}{3} \sec^3 x - \sec x + C$$

$$6). \int \sec^6 x \, dx \quad \sec^4 x = (\sec^2 x)^2 = (1 + \tan^2 x)^2$$

$$= \int \sec^4 x \cdot \sec^2 x \, dx = 1 + 2\tan^2 x + \tan^4 x$$

$$= \int (1 + 2\tan^2 x + \tan^4 x) \cdot \sec^2 x \, dx$$

$$= \int (1 + 2u^2 + u^4) \, du \quad \begin{array}{l} u = \tan x \\ du = \sec^2 x \, dx \end{array}$$

$$= u + \frac{2}{3} u^3 + \frac{1}{5} u^5 + C$$

$$= \tan x + \frac{2}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C$$

$$7). \int \frac{\sin x}{\cos^3 x} \, dx$$

$$u = \cos x$$

$$du = -\sin x \, dx$$

$$= - \int \frac{1}{u^3} \, du$$

$$= - \int u^{-3} \, du$$

$$= - \left(-\frac{1}{2} \right) u^{-2} + C$$

$$= \frac{1}{2u^2} + C$$

$$= \frac{1}{2\cos^2 x} + C$$

$$8). \int_0^{\pi/4} \sec^4 \theta \tan^4 \theta \, d\theta$$

$$= \int_0^{\pi/4} (\sec^2 \theta \tan^2 \theta) \cdot \sec^2 \theta \, d\theta$$

$$= \int_0^{\pi/4} (\tan^2 \theta + 1) \tan^2 \theta \cdot \sec^2 \theta \, d\theta$$

$$= \int_0^1 (u^2 + 1) u^2 \, du$$

$$= \int_0^1 (u^4 + u^2) \, du$$

$$= \left(\frac{1}{5} u^5 + \frac{1}{3} u^3 \right) \Big|_0^1$$

$$= \frac{1}{5} + \frac{1}{3} = \frac{12}{35}$$

$$u = \tan \theta$$

$$du = \sec^2 \theta \, d\theta$$

$$\text{when } \theta = 0: u = 0$$

$$\text{when } \theta = \frac{\pi}{4}: u = 1$$

$$9). \int \sin(5\theta) \sin \theta d\theta$$

$$= \frac{1}{2} \int [\cos(5\theta - \theta) - \cos(5\theta + \theta)] d\theta$$

$$= \frac{1}{2} \int [\cos(4\theta) - \cos(6\theta)] d\theta$$

$$= \frac{1}{2} \left(\frac{1}{4} \sin(4\theta) - \frac{1}{6} \sin(6\theta) \right) + C$$

$$= \frac{1}{8} \sin(4\theta) - \frac{1}{12} \sin(6\theta) + C$$

$$10). \int \sin(8x) \cos(5x) dx$$

$$= \frac{1}{2} \int [\sin(8x + 5x) + \sin(8x - 5x)] dx$$

$$= \frac{1}{2} \int [\sin(13x) + \sin(3x)] dx$$

$$= \frac{1}{2} \left(\frac{-1}{13} \cos(13x) - \frac{1}{3} \cos(3x) \right) + C$$

$$= -\frac{1}{26} \cos(13x) - \frac{1}{6} \cos(3x) + C$$